
Software Requirements Specification

for

Lab/Classroom/ Audi/ Hall Booking system

Version 1.0

**Prepared by
GROUP 12**

Varun Y - B220071CS

Mohammed Asif Hasan - B220397CS

Krishna Harsha - B220042CS

Sai Charan - B220996CS

Hari Vamshi - B220276CS

NIT CALICUT

13th January 2025

Table of Contents

Table of Contents.....	2
Revision History.....	2
1. Introduction.....	3
1.1 Purpose.....	3
1.2 Document Conventions.....	3
1.3 Intended Audience and Reading Suggestions.....	3
1.4 Product Scope.....	3
1.5 References.....	3
2. Overall Description.....	4
2.1 Product Perspective.....	4
2.2 Product Functions.....	4
2.3 User Classes and Characteristics.....	4
2.4 Operating Environment.....	5
2.5 Design and Implementation Constraints.....	5
2.6 User Documentation.....	5
2.7 Assumptions and Dependencies.....	5
3. External Interface Requirements.....	6
3.1 User Interfaces.....	6
3.2 Hardware Interfaces.....	6
3.3 Software Interfaces.....	6
3.4 Communications Interfaces.....	6
4. System Features.....	7
4.1 Authentication and Authorization.....	7
4.2 Room Browsing and Booking.....	7
4.3 Hierarchical Booking Approval.....	8
4.4 Live Room Status Management.....	8
4.5 Booking Timeout and HOLD Mechanism.....	9
4.6 Booking History and Cancellation.....	9
5. Use Cases.....	10
6. Other Nonfunctional Requirements.....	11
6.1 Performance Requirements.....	11
6.2 Safety Requirements.....	12
6.3 Security Requirements.....	12
6.4 Software Quality Attributes.....	12
6.5 Business Rules.....	12
Appendix A: Glossary.....	12

Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

*This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of the **Classroom, Laboratory, and Hall Booking System**. The purpose of this document is to provide a detailed description of system behavior, constraints, and interactions for use by developers, testers, and evaluators.*

1.2 Document Conventions

- *The term “**shall**” indicates a mandatory requirement.*
- *Functional requirements are labeled as **FR-x**.*
- *Non-functional requirements are labeled as **NFR-x**.*
- *All requirements are assumed to be mandatory unless otherwise specified.*

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- *Software developers*
- *Project supervisors*
- *Software testers*
- *Academic evaluators*

Readers should begin with Sections 1 and 2 for an overview and proceed to Sections 4 and 5 for detailed system behavior.

1.4 Product Scope

The Classroom, Laboratory, and Hall Booking System is a web-based application designed to streamline the process of booking institutional resources. It eliminates scheduling conflicts, ensures controlled access through hierarchical approvals, and provides real-time visibility of room usage.

1.5 References

- *IEEE Std 830-1998 – Software Requirements Specification*
- *Spring Boot Documentation*
- *MySQL Database Documentation*

2. Overall Description

2.1 Product Perspective

The system is a **new standalone application** intended to replace manual or semi-digital booking methods currently used in academic institutions. It follows a client-server architecture with a React frontend communicating with a Spring Boot backend, which interacts with an MySQL database.

2.2 Product Functions

The system provides the following major functions:

- Secure authentication using institute email IDs
- Department-wise listing of rooms, labs, and halls
- Time-slot based booking requests
- Hierarchical approval of bookings
- Live status tracking of room usage
- Booking cancellation and timeout handling

2.3 User Classes and Characteristics

User Class	Description
<i>Student</i>	<i>Requests rooms for academic activities</i>
<i>Faculty</i>	<i>Schedules lectures, labs, and events</i>
<i>Room Admin</i>	<i>Approves bookings for assigned rooms</i>
<i>System Admin</i>	<i>Manages room data and system settings</i>

2.4 Operating Environment

Client: Web browsers (Chrome, Firefox)

Frontend: React

Backend: Spring Boot

Database: MySQL DB

Server OS: Linux / Windows

2.5 Design and Implementation Constraints

- *The backend shall be implemented using Spring Boot.*
- *The frontend shall be developed using React.*
- *MySQL Database shall be used for data storage.*
- *Authentication shall be restricted to institute email domains.*

2.6 User Documentation

- *User manual*
- *Admin manual*
- *Online help documentation*

2.7 Assumptions and Dependencies

- *Users have internet connectivity.*
- *Institute email services are operational.*
- *Room administrators are pre-assigned.*
- *Time slots are predefined by the institution.*

3. External Interface Requirements

3.1 User Interfaces

- *Web-based graphical interface*
- *Role-based dashboards*
- *Calendar-based booking view*
- *Status indicators for bookings*

3.2 Hardware Interfaces

- *No specific hardware interfaces required.*

3.3 Software Interfaces

- *MySQL DB for data persistence*
- *Email service for authentication and notifications*

3.4 Communications Interfaces

- *HTTP/HTTPS protocols*
- *Secure RESTful API communication*

4. System Features

4.1 Authentication and Authorization

4.1.1 Description and Priority

This feature enables secure user registration and login using institute-provided email IDs and enforces role-based access to system functionalities.

Priority: High

4.1.2 Stimulus/Response Sequences

1. *User enters institute email ID and password.*
2. *System validates credentials and email domain.*
3. *If valid, system grants access based on the user role.*
4. *If invalid, system displays an authentication error.*

4.1.3 Functional Requirements

REQ-1: The system shall allow users to register and authenticate using institute email IDs only.

REQ-2: The system shall deny access to users with non-institute email addresses.

REQ-3: The system shall enforce role-based access after successful authentication.

4.2 Room Browsing and Booking

4.2.1 Description and Priority

This feature allows users to view available classrooms, laboratories, and halls and request bookings for specific time slots.

Priority: High

4.2.2 Stimulus/Response Sequences

1. *User selects department.*
2. *System displays rooms associated with the selected department.*
3. *User selects room and time slot.*
4. *System checks availability and submits booking request.*

4.2.3 Functional Requirements

REQ-4: The system shall display available rooms based on the user's department.

REQ-5: The system shall allow users to request bookings for specific rooms and time slots.

REQ-6: The system shall prevent multiple bookings for the same room and time slot.

REQ-7: The system shall display booking status as pending, approved, or rejected.

4.3 Hierarchical Booking Approval

4.3.1 Description and Priority

This feature ensures that booking requests follow a hierarchical approval process with designated administrators for each room.

Priority: High

4.3.2 Stimulus/Response Sequences

1. *User submits booking request.*
2. *System forwards request to respective room administrator.*
3. *Administrator approves or rejects the request.*
4. *System updates booking status and notifies the user.*

4.3.3 Functional Requirements

REQ-8: The system shall assign a designated administrator to each room, laboratory, or hall.

REQ-9: The system shall forward booking requests to the respective room administrator.

REQ-10: Administrators shall be able to approve or reject booking requests

4.4 Booking Timeout and HOLD Mechanism

4.4.1 Description and Priority

This feature manages unattended bookings by introducing a cooldown period and a HOLD option.

Priority: Medium

4.4.2 Stimulus/Response Sequences

1. *Booking start time is reached.*
2. *User is not present in the room.*
3. *System starts a 30-minute cooldown timer.*
4. *User clicks HOLD or booking is cancelled after timeout.*

4.4.3 Functional Requirements

REQ-13: The system shall initiate a 30-minute cooldown period if the user is not present at booking start time.

REQ-14: The system shall cancel the booking after cooldown expiry.

REQ-15: The system shall allow users to retain bookings by clicking the HOLD button during the cooldown period.

4.5 Booking History and Cancellation

4.5.1 Description and Priority

This feature allows users to view previous and upcoming bookings and cancel bookings when required.

Priority: Medium

4.5.2 Stimulus/Response Sequences

1. *User accesses booking dashboard.*
2. *System displays booking history.*
3. *User selects a booking to cancel.*
4. *System updates booking status.*

4.5.3 Functional Requirements

REQ-16: The system shall allow users to view past and upcoming bookings.

REQ-17: The system shall allow users to cancel bookings.

5. Use Cases

UC-01: User Login

Actor: Student / Faculty / Admin

Precondition: User has an institute email ID

Main Flow:

1. User enters email and password
2. System validates credentials
3. User is logged into the system

Postcondition: User session is created

UC-02: View Available Rooms

Actor: User

Precondition: User is authenticated

Main Flow:

1. User selects department
2. System displays available rooms and time slots
Postcondition: Room list is displayed

UC-03: Request Room Booking

Actor: Faculty / Student

Precondition: User is authenticated

Main Flow:

1. User selects room and time slot
2. System checks availability
3. Booking request is submitted
Postcondition: Booking status set to pending

UC-04: Approve or Reject Booking

Actor: Room Admin

Precondition: Booking request exists

Main Flow:

1. Admin reviews booking request
2. Admin approves or rejects request
Postcondition: Booking status updated

UC-05: Hold Booking

Actor: User

Precondition: Cooldown period is active

Main Flow:

1. User clicks HOLD button
2. System retains booking
Postcondition: Booking remains active

UC-06: Cancel Booking

Actor: User

Precondition: Booking exists

Main Flow:

1. User selects booking
2. User cancels booking
Postcondition: Booking is cancelled

UC-07: View Booking History

Actor: User

Precondition: User is authenticated

Main Flow:

1. User views booking dashboard
 2. System displays past and upcoming bookings
- Postcondition:** Booking history displayed

6. Other Nonfunctional Requirements

6.1 Performance Requirements

***NFR-1:** Booking availability checks shall respond within 2 seconds.*

***NFR-2:** The system shall support concurrent users without degradation.*

6.2 Safety Requirements

- *No safety-critical requirements applicable.*

6.3 Security Requirements

NFR-3: *All users shall be authenticated.*

NFR-4: *Role-based access control shall be enforced.*

6.4 Software Quality Attributes

- *Usability*
- *Reliability*
- *Maintainability*
- *Scalability*

6.5 Business Rules

- *Only authorized administrators may approve bookings.*
- *Users may only book rooms permitted by their role and department.*

Appendix A: Glossary

- **Cooldown:** *Grace period before automatic booking cancellation*
- **HOLD:** *Option to retain booking during cooldow*

